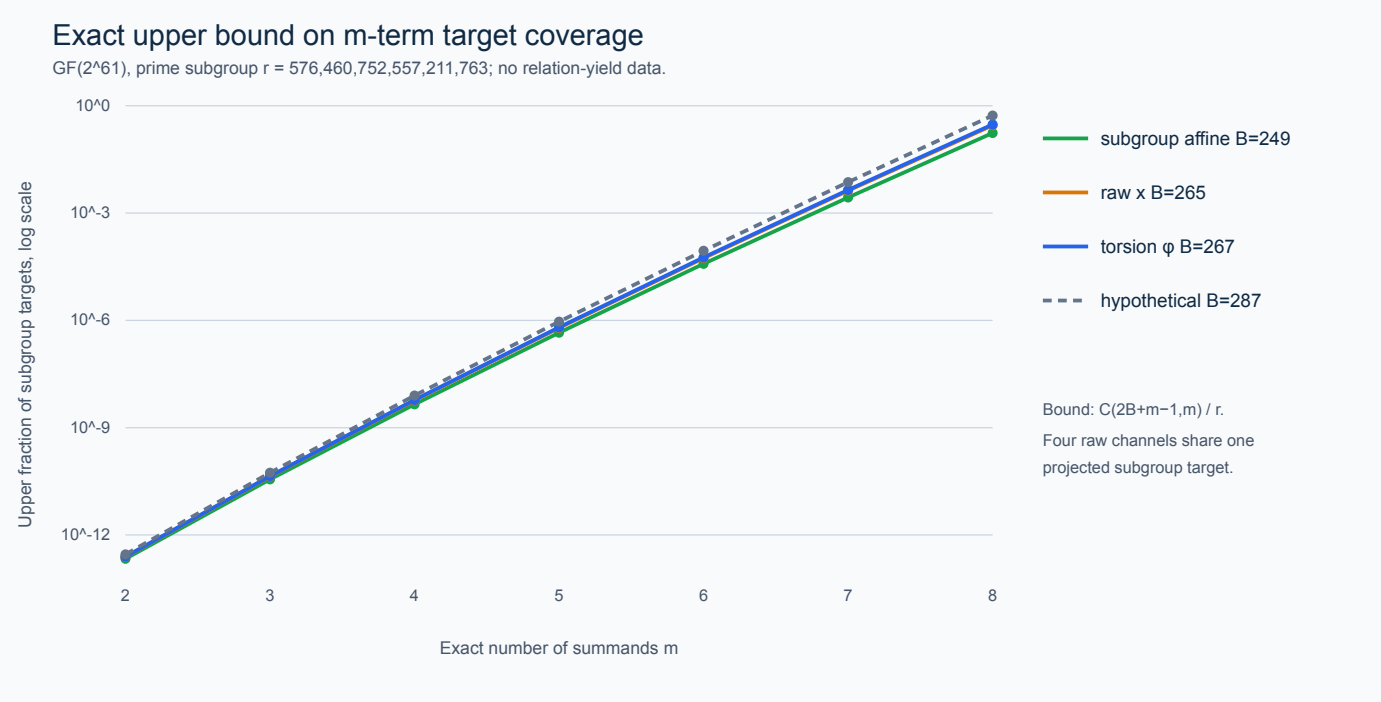
Signed-multiset coverage ceilings for binary m=2–8

17 October 2026 · certified prime subgroup order 576,460,752,557,211,763

Question. What fraction of uniform subgroup targets could an ideal m-term oracle represent from the prepared bases, before solver or rank costs?



Exact bound. For B sign-folded subgroup columns, unordered length-m multisets from their 2B signed values number $\binom{2B+m-1}{m}$. Distinct group sums are no more numerous. Thus the uniform-target coverage fraction is at most $\min\left(1, \frac{\binom{2B+m-1}{m}}{r}\right)$. Group collisions, cancellations, invalid lifts, solver failure and duplicate rows can only reduce useful yield.

m=8 values. Prepared B=249 direct: 17.2124%; B=265 raw x: 28.2327%; B=267 torsion φ: 29.9713%. A hypothetical B=287 reference gives 53.2229%. At m=7, B=267 has an upper fraction of 0.4432%. These are derived ceilings, not measured relation success.

Channel accounting. All four raw channels project under the cofactor map to one subgroup target. Solving all four is needed for complete raw-lift coverage; their projected target coverage is not multiplied by four.

Limit. No solver work, degree of regularity, relation yield, useful rank, complete index-calculus cost or speed improvement has been measured. Source and exact integer counts: `research/icperf_binary_coverage_bound_20261017/bound.json`.